The Anatomy of Factor Crowding: Volatility and Business-Cycle Risk*

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Abstract

Investors watch returns-based crowding scores to see where factor diversification may fail. Using 122 equity factors in the US and other developed markets, we show that the standard score is mostly a volatility ranking (rank correlation 0.96): factors it flags as crowded fall more in recessions because they are more volatile. Pointed at the business cycle, the same decomposition adds information, beyond volatility, about which factors' returns will be most sensitive to the economy over the next two years. Neither score is a profitable return signal, and crowding caps add nothing to risk budgets that account for correlations.

Keywords: Factor Investing, Crowding, Centrality, Target PCA, Business Cycle, Risk Budgeting.

JEL codes: G11, G12, E32

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1 Introduction

Multi-factor portfolios diversify across value, profitability, investment, and low-risk premia, but the risk they diversify is not constant. Over a century of data, the risk of individual factors and of diversified factor portfolios varies substantially over time, and diversification across factors does not neutralize it (Ilmanen, Israel, Lee, Moskowitz, and Thapar, 2021). Investors therefore monitor where the common risk of a factor portfolio accumulates, and many institutions maintain committees devoted to monitoring crowding (Chincarini, Lazo-Paz, and Moneta, 2026). Crowded trades, however, are difficult to identify (Pojarliev and Levich, 2011), and positions are observed only for subsets of investors. The most widely used crowding scores are therefore inferred from returns: the absorption ratio and the centrality measure derived from it attribute the common variation of a set of assets to each asset and are used to locate crowded sectors and factors (Kritzman, Li, Page, and Rigobon, 2011; Kinlaw, Kritzman, and Turkington, 2019).

The business cycle is the most important source of such common risk, and it matters for factor investors not only through risk. Exhibit 1 shows how the risk-adjusted performance of the 122 US equity factors studied in this article changes across the four phases of a real-time business-cycle indicator. Relative to its average level, the appraisal ratio of the typical factor is lower in expansions (-0.10) and higher in recessions ($+0.16$) and recoveries ($+0.12$), and the dispersion across factors widens in the downturn phases. For a quarter of the factors (31 of 122), equal performance across phases is rejected at the 5% level, more often than when the sequence of phases is shifted in time (placebo p-value of 0.04). Which factors carry a portfolio's business-cycle exposure is therefore a first-order question for a multi-factor investor, and crowding scores are the tool many investors use to answer it.

What these scores measure is less clear than how they are used. They are built from the covariance matrix of returns, and by design they rate more volatile assets as more central:

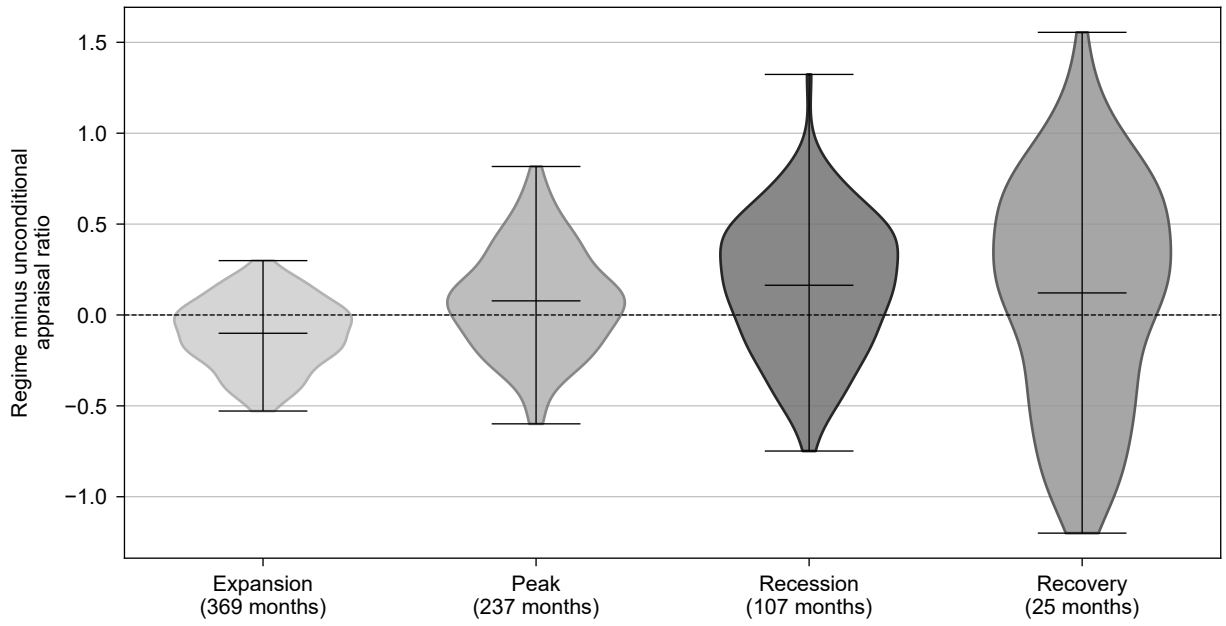


Exhibit 1: **Factor Performance across the Business Cycle.** Distribution across the 122 JKP US factors of the regime-conditional minus the unconditional appraisal ratio, July 1963–December 2024. For each factor, monthly returns are regressed on four regime dummies (no constant) and the five Fama–French factors (Fama and French, 2015); the regime appraisal ratio is the annualized regime alpha over the annualized residual volatility. Regimes of the real-time business-cycle indicator of Swade et al. (2024): Expansion (positive and improving), Peak (positive and deteriorating), Recession (negative and deteriorating), Recovery (negative and improving). Horizontal bars: mean, minimum, and maximum.

their authors use volatility as a proxy for an asset's vulnerability (Kinlaw, Kritzman, and Turkington, 2012) and read high volatility as a symptom of crowding, which raises an asset's volatility (Kinlaw et al., 2019). Holdings data point the other way. Capital in factor strategies tends to leave when their volatility rises (Hanson and Sunderam, 2014), institutions use volatility as a signal to exit a strategy rather than causing it through their crowding, and a returns-based crowding proxy predicts tail risk for reasons unrelated to institutional crowding (Barroso, Edelen, and Karehnke, 2022). Where returns-based crowding measures have been validated, they are built from comovement free of scale and tested against volatility, for single strategies such as momentum and beta arbitrage (Lou and Polk, 2022; Huang, Liu, Lou, and Polk, 2024).

For the centrality score across factors, the measure most directly relevant to a multi-factor allocator, four questions remain open. Is it more than a volatility ranking? Construction

alone does not settle this: the leading components of a covariance matrix favor volatile series, but a score could still carry information about common exposure beyond volatility, and its authors read high centrality as evidence of crowding. Does its information survive out of sample, across data sets and regions? Can it be made informative about the risk that matters most for a factor portfolio, its exposure to the business cycle? And does it help an investor who already sizes factor sleeves by risk? We are not aware of a study that answers them. This article does, with two contributions, each tied to a portfolio decision: where a factor portfolio's business-cycle risk is concentrated, and how much risk each factor sleeve should carry.

What crowding scores measure. Across the 122 capped value-weighted factors of Jensen, Kelly, and Pedersen (2023) that are not built from past returns, size, or trading activity, standard centrality ranks factors almost exactly as trailing volatility does: the cross-sectional rank correlation is 0.96 in the United States and 0.95 in developed markets outside the United States. In all ten US recessions since 1957, the most crowded factors suffered deeper drawdowns than the least crowded, by 6.6 percentage points on average, and a plain volatility sort delivers the same difference (6.8 points). The same holds in the nine market sell-offs of at least 15% outside recessions, where crowded positions should be most visible. Computing the score from correlations weakens the link to volatility (rank correlation of 0.57) but adds no information about future business-cycle exposure beyond volatility and other simple controls.

How to point crowding at business-cycle risk. Target principal component analysis (Bryzgalova, DeMiguel, Li, and Pelger, 2026) directs the decomposition toward economic information. We center it and use a single real-time innovation of a business-cycle indicator as the target. The resulting macro-targeted score adds information about which factors' returns will be most sensitive to the business cycle over the next two years, beyond volatility, market beta, past cycle sensitivity, and standard crowding. The result holds in the United States (t-statistic of 3.3, 1968–2024) and out of sample in developed markets outside the United States (2.4; pooled, 3.1), and the pooled result survives a control for a long-window

cycle beta (2.6). The effect is moderate: moving a factor from the bottom to the top of the ranking raises its expected rank in future cycle sensitivity by about eight percentiles, and the factors it ranks highest moved together in every US recession since 1957 (average pairwise correlation of 0.22, against 0.04 for the lowest-ranked). The score is an adjustment to a ranking by volatility and cycle beta, not a substitute for it: on its own it is only weakly related to future cycle sensitivity in the United States and negatively related in developed markets. The original, uncentered design leaves the crowding map almost unchanged, and its apparent value comes from past premia, as the method's own theory predicts for an uninformative target.

Neither score is a profitable return signal. Tilting toward crowded factors earns less than the additional risk implies (an annual alpha of -0.13% against the equal-weighted factor portfolio, t of -4.8). Crowding caps help portfolios that ignore correlations, mainly equal-weighted ones, because they remove weight from volatile factors; they add nothing to risk budgets that account for correlations.

In plain terms: when a standard crowding report says that a factor is crowded, it mostly says that the factor is volatile, which the risk model already shows. Such a report is not wrong, but it adds little. Pointed at the business cycle, the same calculation becomes useful: beyond volatility and the standard report, it adds information about which factor sleeves are likely to move most with the economy over the next two years. For a chief investment officer, this implies a division of labor. Risk budgets should decide how large each factor sleeve is; a score directed at the business cycle should show where the portfolio's exposure to the economy is concentrated and which sleeves to stress in a downturn scenario. Neither tool should be used to forecast returns.

The article contributes to four strands of work. First, studies of returns-based crowding propose the absorption ratio and centrality (Kritzman et al., 2011; Kinlaw et al., 2012, 2019) and relate crowding measured from comovement to subsequent factor returns (Baltas, 2019);

we show that the factor-level centrality score is, in its standard form, a volatility ranking, and how to make it informative about the business cycle. In this respect the article follows recent evidence in this journal that a widely watched market indicator need not carry the risk investors attribute to it (Kritzman and Turkington, 2026). Second, studies of positioning measure crowding from short interest, holdings, and order flow (Hanson and Sunderam, 2014; Brown, Howard, and Lundblad, 2022; Barroso et al., 2022; Chincarini et al., 2026); our scores are returns-based, and our results explain why a volatility-dominated returns-based score cannot stand in for positioning. Third, target PCA (Bryzgalova et al., 2026) was developed to recover weak pricing factors; we apply it to a risk-monitoring object and show which design choices make the target informative. Fourth, work on factor allocation shows that risk-based combinations of factors improve outcomes (Clarke, de Silva, and Thorley, 2020); we show that crowding caps add nothing to risk budgets that account for correlations.

2 Measuring Crowding: Scores, Data, and Design

A crowding score of this kind starts from a principal component decomposition of the trailing returns of N_t factors. Each component carries a share of the common variation, and each factor loads on each component. The centrality of factor i is its share of the leading components, weighted by the variance they explain (Kinlaw et al., 2019):

$$C_i \;=\; \frac{\sum_{j=1}^{K} \sigma_j^2 \cdot |v_i^j| \big/ \sum_{k=1}^{N_t} |v_k^j|}{\sum_{j=1}^{K} \sigma_j^2}, \tag{1}$$

where σ_j^2 is the variance of component j and v_i^j the loading of factor i on it. The scores are nonnegative and sum to one, so they divide the common variation among the factors. Because the loadings enter in absolute value, centrality measures the size of a factor's exposure to the common components, not its direction.

Computed from the covariance matrix, as in Kinlaw et al. (2019), centrality mixes two

properties: how strongly a factor moves with the others and how volatile it is. The leading components of a covariance matrix load most heavily on the most volatile series, so a volatile factor is central even if its comovement with the others is ordinary. Computing the same score from the correlation matrix removes the scale; we denote it C^{COR} and the standard score C^{PCA}. Comovement measured free of scale is the quantity that comovement-based crowding studies validate (Pollet and Wilson, 2010; Lou and Polk, 2022; Huang et al., 2024).

We use *crowding* in the conventional, returns-based sense of this literature: a score that measures how much of a factor's risk is common, not who holds it. Positions are observed only for subsets of investors (Brown et al., 2022), and common ownership is one of several sources of common return variation (Antón and Polk, 2014). Because capital tends to leave factor strategies when their volatility rises (Hanson and Sunderam, 2014; Barroso et al., 2022), a score dominated by volatility should not be read as a measure of positioning.

Target PCA (Bryzgalova et al., 2026) adds penalties to the principal component problem that pull the components toward economic information: a cross-sectional target favors directions in which factors' loadings on macroeconomic shocks line up, and a time-series target favors months that carry information about the shocks. The method's theory implies that a target helps only if it is informative about the factors' loadings; an uninformative target leaves the decomposition unchanged (Bryzgalova et al., 2026, pp. 19–20). Two design choices therefore matter. First, the original estimator works with the uncentered second moment of returns, as risk-premium PCA does (Lettau and Pelger, 2020), so its components also reflect average returns; we center returns so that any change in the map comes from the target. Second, macroeconomic series explain little of factor returns (Ilmanen et al., 2021), so a target built from several noisy macroeconomic components estimated over a five-year window carries little information. We use one real-time innovation of a business-cycle indicator, with loadings estimated over all available history. The resulting macro-targeted centrality minus standard centrality, $M2$, measures how much the business-cycle target shifts a factor's share of the common variation.

Each month, the macro-targeted score is computed in four steps from information available at that date. First, the trailing 60 months of factor returns are demeaned, so that the decomposition reflects comovement and not average returns. Second, the cross-sectional target is each factor's loading λ_i on the innovation of the business-cycle indicator, the part of its monthly change that a first-order autoregression does not predict; the loading is estimated from all available history up to three months before the formation date. Third, the time-series target emphasizes the months in which the indicator deteriorated over the previous year. Fourth, principal component analysis is applied to the targeted operator

$$\Omega_t \;=\; \frac{1}{NT}\left(I_N + \tilde{\gamma}_{XS} P_\lambda\right) \tilde{R}_t^\top \left(I_T + \gamma_{TS} P_D\right) \tilde{R}_t \left(I_N + \tilde{\gamma}_{XS} P_\lambda\right), \tag{2}$$

where $\tilde{R}_t$ is the demeaned $T \times N$ return matrix, P_λ and P_D project on the cross-sectional target (the loadings λ) and on the time-series target (the downturn series D), and γ_{XS} and γ_{TS} set the strength of each target; $\tilde{\gamma}_{XS}$ is γ_{XS} rescaled for the number of factors (Section B of the Appendix; Bryzgalova et al., 2026). Without targets, Ω_t is the covariance matrix and Equation (1) returns standard centrality. With targets, Equation (1) applied to Ω_t gives the macro-targeted centrality $C^{\text{XS-TS},c}$, and

$$M2_{i,t} \;=\; C_{i,t}^{\text{XS-TS},c} - C_{i,t}^{\text{PCA}} \tag{3}$$

is the shift in factor i's share of the common variation that the business-cycle target causes. Because both centralities sum to one, $M2$ sums to zero across factors: the target moves weight from factors whose comovement is unrelated to the cycle toward factors whose comovement is cyclical. The absolute loading $|\lambda_i|$ is itself a long-window estimate of the factor's cycle beta and serves as a control.

A crowding score is useful for risk monitoring if it tells an investor today which factors will be most exposed to the business cycle in the future. For each factor i and month t, we regress its returns over the next 24 months on the monthly change in the business-cycle

indicator,

$$r_{i,s} \;=\; a_i + b_i\,\Delta Z_s + \varepsilon_{i,s}, \qquad s = t+1, \ldots, t+24, \tag{4}$$

and record two outcomes: the factor's future cycle sensitivity $|b_i|$, written $|\beta(\Delta Z)|$, and the cycle's share of the factor's variance, the R^2 of the regression, written $R^2(\Delta Z)$. The first scales with the factor's volatility; the second does not. Each month, we relate the rank of each score to the rank of the outcome across factors, alone (a rank correlation) and in a regression that also includes the ranks of simpler measures an investor already has: trailing volatility, the absolute beta on the regional equity market, the factor's past cycle sensitivity, and the size of its loadings on the macroeconomic components of the indicator. The coefficient on the score in this regression measures the information the score adds. For the United States, these incremental tests start in 1968, the window fixed in advance to match the development sample, in which the market-beta control is available from that year. Monthly coefficients are averaged over time, with Newey–West standard errors that account for the overlapping 24-month outcome windows (Fama and MacBeth, 1973; Newey and West, 1987). Rank-based statistics are robust to the outliers that can dominate least-squares cross-sectional regressions.

This design lets the data answer the questions of the introduction directly. If the standard score is built largely from volatility, its ranking should follow volatility, whereas centrality computed from correlations should not. A score that measures cyclical comovement should identify the factors whose returns will be most sensitive to the business cycle, beyond their volatility and their standard crowding score. And if crowding scores measure risk rather than mispricing, tilting toward crowded factors should earn no risk-adjusted premium, and the scores should add nothing to risk-based weights.

The main data are the factor returns of Jensen et al. (2023) (JKP), a public database that rebuilds 153 published equity factors consistently for 93 countries. Each factor is a long–short portfolio of the top and bottom terciles of a firm characteristic, signed as in the

original study. Breakpoints are set on stocks larger than the 20th NYSE size percentile, so that the terciles are not dominated by microcaps, and stocks are weighted by their market capitalization capped at the 80th NYSE percentile, so that tiny stocks have tiny weights and no single large stock dominates a portfolio. These choices make the factors closer to the strategies institutional investors can implement than the equal-weighted portfolios of many original studies, and the same construction in every region makes the regional results comparable.

We exclude 31 of the 153 factors: those whose sorting variable is built from past returns (momentum, reversal, and seasonality), from the price level or size, or from trading activity (liquidity, turnover, and volume). There are two reasons. Centrality is itself computed from past returns, so its relation to factors sorted on past returns would be partly mechanical; Kinlaw et al. (2019) exclude momentum from their analysis of crowded factors for the same reason. And these signals reflect trading and microstructure effects rather than the fundamental exposures through which the business cycle reprices firms; they are also the group of anomalies least likely to replicate (Hou, Xue, and Zhang, 2020). Risk measures computed from returns, such as betas and volatilities, are kept, because they describe the risk of the firms a factor holds and are the most direct candidates for the volatility explanation we test. Section 6 shows that the volatility result holds with all 153 factors and without the 18 return-based risk measures, and that the macro-targeted score is weaker without them. The remaining 122 factors are listed in Exhibit A2 of the Appendix, and nine of the clusters of Jensen et al. (2023), which contain 109 of them, serve as themes for interpretation (Exhibit A1); the other 13 belong to clusters whose remaining members are excluded and enter only the factor-level tests. The main sample is the United States, with portfolios formed monthly from January 1957 to December 2024 (816 months). Developed markets outside the United States (22 countries, formation from December 1990) are the out-of-sample block, and emerging markets (from January 1994) are reported descriptively.

We measure the business cycle with the real-time indicator of Swade et al. (2024), the

average of five expanding-window standardized components: the price-to-earnings ratio, the negative AAA–BAA credit spread, industrial-production growth, and expected and unexpected inflation from a recursively estimated model of consumer prices. The same US indicator is used for all regions as a measure of the common business cycle. Its monthly change, ΔZ, defines the outcome of Equation (4). NBER recession dates serve as an external benchmark.

All scores are computed each month from the trailing 60 months of returns of the factors with a complete history, with $K = 5$ components. A factor enters the cross-section once it has five years of returns, so the number of US factors grows from 89 in 1957 to 122 by the 1990s. Every input to the macro-targeted score is available at the formation date: the innovation of the indicator and the factors' loadings on it are estimated on data up to three months before the formation date, and the time-series target excludes the last three months of the window, the publication lag of the macroeconomic inputs. The target strengths are those of the original design used in the development sample ($\gamma_{XS} = 0.1$, $\gamma_{TS} = 10$) and are not tuned; the estimator is documented in Section B of the Appendix. The business-cycle regimes of the indicator are used only to describe results, never in a score or a portfolio. The indicator is real-time in its construction, with expanding-window standardization and recursive estimation, but it uses current data vintages; of its inputs, industrial production in particular is revised after publication.

The analysis was first developed on the Open Source Asset Pricing (OSAP) anomaly portfolios of Chen and Zimmermann (2022), aggregated into 27 economic themes (Supplemental Material). The tests on the JKP data, including the design of the macro-targeted score, were specified in advance in a dated research log, so the JKP data are a confirmatory sample for these tests. Two steps came after the first JKP results and are reported as such: the control for a long-window cycle beta (Section 3) was added post hoc to address an alternative explanation, and the JKP data became the main data set after these results were known. The log is part of the replication package. Section 6 reports where the two data sets

differ.

3 What Crowding Scores Capture

Exhibit 2 shows the crowding map that a risk report based on standard centrality would display, and how closely it follows volatility.

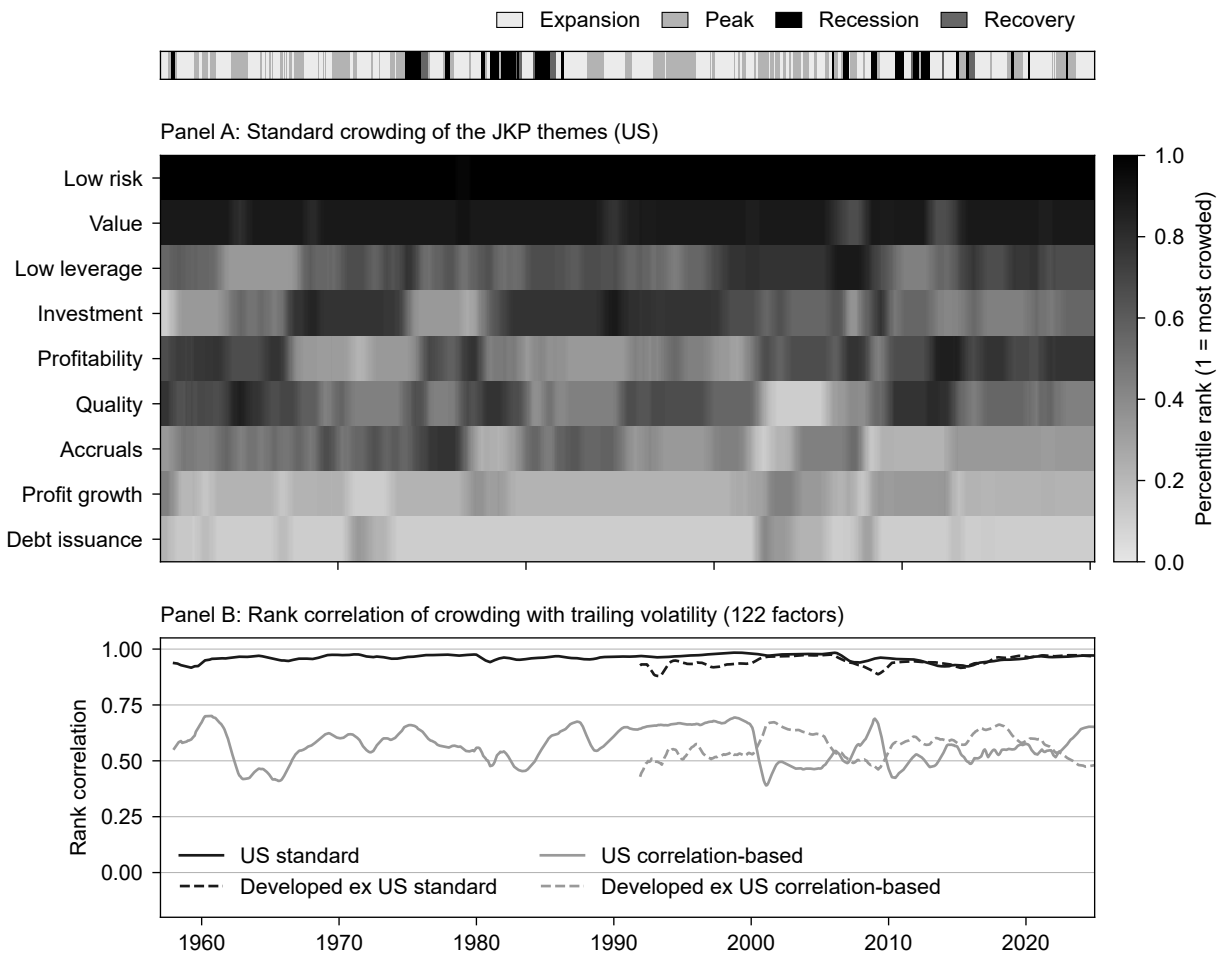


Exhibit 2: **The Crowding Map and Its Volatility Content.** Panel A: monthly cross-sectional percentile rank of standard centrality C^{PCA} of the nine JKP themes (US, $K = 3$), 12-month trailing mean; strip: regime of the real-time business-cycle indicator. Panel B: cross-sectional rank correlation of standard centrality and of correlation-based centrality with trailing 60-month volatility across the 122 JKP factors, United States and developed ex US, 12-month trailing mean.

The map is persistent. Low-risk and value factors are the most crowded themes throughout, and debt issuance and profit growth the least crowded. Panel B shows why: across the

122 factors, the rank correlation between standard centrality and trailing volatility stays near 0.95 in every decade and in both regions. The low-risk and value themes are the most volatile themes in the sample (Exhibit A1). The theme a crowding report would flag as the most crowded in nearly every month since 1957 is simply the most volatile one. The result does not depend on the factors that sort stocks on statistics of their past returns: without the 18 factors built from betas, volatilities, maximum returns, and skewness, the rank correlation is unchanged (0.96 in the United States and 0.95 in developed markets; Exhibit 9).

Exhibit 3 quantifies this and asks which score carries information about future business-cycle exposure.

Standard centrality has an average rank correlation with volatility of 0.96 in the United States, 0.95 in developed markets, and 0.90 in emerging markets; in the development sample of anomaly themes it is 0.92 (Panel A). It is strongly related to future cycle sensitivity (rank correlation of 0.38, t of 14.9), but no more than volatility itself: once volatility, market beta, past cycle sensitivity, and the factors' macroeconomic loadings are controlled for, its coefficient is -0.11 (t of -1.8) in the United States and insignificant in developed markets (Panel B). For the scale-free outcome, the cycle's share of a factor's variance, standard centrality carries no information (Panel C).

Correlation-based centrality removes part of the scale (rank correlation with volatility of 0.57) and is still related to future cycle sensitivity (0.21, t of 11.6), but it adds no information beyond the controls and standard centrality, in either region or for either outcome. Two factors with the same volatility do not differ predictably in their future cycle exposure according to how much of their risk is common. Standard crowding scores therefore tell an investor which factors are volatile, a quantity the risk model already reports.

Pointing the decomposition at the business cycle changes the map: the rank correlation of the macro-targeted score with volatility is 0.77 in the United States and 0.65 in developed markets (Exhibit 3, Panel A), against 0.96 and 0.95 for standard centrality. The shift $M2$

Panel A: Mean rank correlation with trailing volatility

	US	Developed ex US	Emerging	OSAP themes
C^{PCA}	0.961	0.945	0.905	0.921
C^{COR}	0.567	0.561	0.436	0.255
$C^{\text{XS-TS},c}$	0.765	0.650	0.615	

Panel B: Future cycle sensitivity $|\beta(\Delta Z)|$, next 24 months

	Spearman	Incremental, all controls		
Signal	US	US 1968–2024	Developed ex US	Pooled
C^{PCA}	0.382 (14.9)	−0.106 (−1.8)	0.073 (1.1)	−0.037 (−0.7)
C^{COR} (also C^{PCA})	0.211 (11.6)	−0.007 (−0.3)	0.020 (0.8)	−0.017 (−1.0)
$M2$ (also C^{PCA})	0.056 (2.3)	0.084 (3.3)	0.052 (2.4)	0.078 (3.1)
$M2$ (also C^{PCA}, $\|\lambda\|$)		0.092 (3.1)	0.061 (1.7)	0.086 (2.6)
$\|\lambda\|$ (also C^{PCA})	0.279 (12.1)	0.081 (3.3)	0.036 (2.1)	0.066 (2.9)

Panel C: Future cycle share of variance $R^2(\Delta Z)$, next 24 months

	Spearman	Incremental, all controls		
Signal	US	US 1968–2024	Developed ex US	Pooled
C^{PCA}	0.000 (0.0)	−0.083 (−1.5)	0.086 (1.2)	−0.025 (−0.4)
C^{COR} (also C^{PCA})	−0.015 (−0.9)	−0.014 (−0.6)	−0.014 (−0.6)	−0.024 (−1.5)
$M2$ (also C^{PCA})	0.070 (2.8)	0.056 (2.0)	0.054 (2.5)	0.057 (2.3)
$M2$ (also C^{PCA}, $\|\lambda\|$)		0.080 (2.3)	0.114 (2.6)	0.096 (2.9)
$\|\lambda\|$ (also C^{PCA})	0.049 (2.2)	0.045 (1.7)	0.018 (0.9)	0.037 (1.5)

Exhibit 3: **What Crowding Measures Capture.** JKP factors (122 factors, capped value weights). Panel A: average monthly cross-sectional rank correlation of each measure with trailing 60-month volatility; C^{PCA}: centrality from the covariance matrix, C^{COR}: from the correlation matrix, $C^{\text{XS-TS},c}$: centered target PCA; OSAP themes: reference from the development sample. Panels B and C: Fama–MacBeth averages of monthly rank statistics relating each signal at month t to each factor's absolute slope (B) and R^2 (C) from regressing its returns on the monthly change of the real-time cycle indicator over $t+1,\ldots,t+24$. Incremental: coefficient of the signal's rank in a regression of the outcome's rank on the ranks of the signal and the controls (trailing 60-month volatility, absolute market beta, trailing $|\beta(\Delta Z)|$ estimated without the last three months, norm of the loadings Λ on the five cycle components, and the measures named after "also" in the row label). Spearman column: rank correlation without controls. $M2 = C^{\text{XS-TS},c} - C^{\text{PCA}}$; $|\lambda|$: expanding-window loading on the cycle innovation. Pooled: ranks demeaned within block, common months. Newey–West t (24 lags) in parentheses. Formation 1957–2024 (US), 1990–2024 (developed ex US), 1994–2024 (emerging).

adds information about which factors' returns will be most sensitive to the business cycle. Controlling for volatility, market beta, past cycle sensitivity, macroeconomic loadings, and standard centrality, its coefficient for future cycle sensitivity is 0.084 (t of 3.3) in the United States and 0.052 (t of 2.4) in developed markets outside the United States, data the design was not developed on; pooled, it is 0.078 (t of 3.1). The same holds for the cycle's share of

a factor's variance, the outcome that does not scale with volatility (pooled 0.057, t of 2.3).

A natural objection is that the target is itself a long-window estimate of each factor's cycle beta, and that $M2$ only passes on that estimate. The long-window beta does carry information about future cycle sensitivity (pooled 0.066, t of 2.9), but adding it as a control leaves $M2$ significant (pooled 0.086, t of 2.6; for the cycle's share of variance, 0.096, t of 2.9). In developed markets alone, the coefficient beyond the long-window beta is 0.061 (t of 1.7) for cycle sensitivity and 0.114 (t of 2.6) for the cycle's share of variance. The targeted decomposition adds information through the covariance structure: a factor whose own cycle beta is small but which moves with the cyclical factors gains centrality under the target, and the long-window beta does not register that.

The design is what makes the target informative. Applied with the original specification, uncentered and with a target built from five macroeconomic components estimated over the 60-month window, target PCA leaves the crowding map almost unchanged (rank correlation of 0.94 to 0.97 with the untargeted decomposition), its macroeconomic component carries no information about future cycle exposure (t-statistics between -1.0 and 1.6), and the higher returns of its signals come from the average-return component of the uncentered second moment (Exhibit A6 of the Appendix). This is the outcome the method's theory predicts when the target is uninformative about the loadings (Bryzgalova et al., 2026).

The effect is moderate in size: a coefficient of 0.08 on ranks means that moving a factor from the bottom to the top of the $M2$ ranking raises its expected rank in future cycle sensitivity by about eight percentiles, holding the controls fixed. On its own, the $M2$ ranking is only weakly related to future cycle sensitivity: its rank correlation is 0.056 in the United States, against 0.28 for the long-window cycle beta and 0.38 for standard centrality, and in developed markets it is negative (-0.064, t of -3.0); its value lies in adjusting a ranking by volatility and cycle beta, not in replacing it. It is a measure of how strongly a factor will be exposed to the business cycle, not of the direction of its next move.

4 Crowding in Recessions and Market Stress

Recessions are the episodes in which a factor portfolio's common exposure matters most. Exhibit 4 sorts the factors on each score in the month before each NBER recession and follows the most and least crowded 30% through it.

		US (10 recessions)				Developed ex US (3)	
Sorted on	Statistic	Top 30%	Bottom 30%	Difference (t)	Share	Difference (t)	Share
C^{PCA}	Maximum drawdown (%)	−10.62	−4.03	−6.59 (−8.1)	100%	−6.27 (−4.6)	100%
	|Cumulative return| (%)	10.03	5.25	4.77 (4.0)	100%	5.15 (2.3)	100%
	Pairwise correlation	0.14	0.13	0.00 (0.1)	60%	−0.02 (−0.5)	33%
Trailing volatility	Maximum drawdown (%)	−10.59	−3.75	−6.84 (−8.1)	100%	−6.49 (−4.0)	100%
	|Cumulative return| (%)	10.08	5.48	4.60 (3.8)	90%	5.29 (2.1)	100%
	Pairwise correlation	0.14	0.13	0.01 (0.2)	50%	0.02 (0.3)	67%
C^{COR}	Maximum drawdown (%)	−7.99	−5.13	−2.86 (−4.9)	100%	−4.79 (−4.3)	100%
	|Cumulative return| (%)	9.80	5.31	4.49 (6.0)	100%	4.19 (3.7)	100%
	Pairwise correlation	0.15	0.10	0.05 (1.0)	50%	0.00 (0.0)	33%
$M2$	Maximum drawdown (%)	−7.56	−7.90	0.34 (0.4)	30%	2.98 (3.3)	0%
	|Cumulative return| (%)	8.28	8.47	−0.19 (−0.1)	50%	0.66 (0.4)	67%
	Pairwise correlation	0.22	0.04	0.18 (4.6)	100%	0.13 (1.1)	67%

Exhibit 4: **Crowded Factors in NBER Recessions.** JKP factors (capped value weights). In the month before each NBER recession (peak to trough) that starts after the first formation month, the factors in the top and bottom 30% of each signal; averages across factors, then across recessions. Maximum drawdown: within-recession maximum drawdown of each factor; |cumulative return|: absolute cumulative log return over the recession; pairwise correlation: mean correlation of the group's monthly returns during the recession. Difference: top minus bottom, t-statistic across recessions (developed ex US: three recessions, descriptive). Share: fraction of recessions with a deeper drawdown, a larger move, or a higher correlation of the top group. US: 1957–2024; developed ex US: 2001, 2007–09, 2020.

In all ten US recessions, the most crowded factors by standard centrality suffered deeper drawdowns than the least crowded (−10.6% against −4.0% on average), and they moved more in either direction. Sorting on trailing volatility gives the same result, recession by recession (Exhibit A3 of the Appendix), and the same holds in the three recessions since 1990 in developed markets. The crowded factors do not move together: their pairwise correlation during recessions is no higher than that of the least crowded factors. Because centrality uses the absolute size of each factor's loadings, factors on opposite sides of the same common component are both central, and the crowded set contains offsetting positions rather than one position.

Sorting on the macro-targeted shift does not identify the factors that lose most in a recession: in the United States, the drawdowns of the top and bottom groups are similar (difference of 0.34 percentage points, t of 0.4), and in developed markets the factors with the largest $M2$ fell less (by 2.98 points). What the sort does identify is comovement: in all ten US recessions, the factors with the largest $M2$ moved together more than those with the smallest (average pairwise correlation of 0.22 against 0.04, t of 4.6), although not significantly so in developed markets (t of 1.1). This is consistent with what $M2$ measures: the strength of a factor's future cycle exposure in either direction, not whether the exposure pays off or costs in a downturn. For a stress test, the crowded set is best described by the signed loadings of its members rather than treated as one position.

If standard centrality measured crowded positions rather than volatility, the difference should be most visible in market sell-offs that are not driven by the business cycle, when crowded investors exit together. We therefore repeat the analysis for every US market drawdown of at least 15% since 1957 that does not overlap an NBER recession, a rule fixed before the analysis. Nine episodes qualify, among them 1987, the 1998 Russian and LTCM crisis, the 2011 euro-area crisis, and 2022, with market drawdowns of 15% to 32% (Exhibit A4 of the Appendix). Exhibit 5 follows the most and least crowded 30% of factors from the market peak.

The result is the same as in recessions. Within the episodes, the most crowded factors suffered deeper drawdowns than the least crowded (a difference of -1.9 percentage points, t of -2.5; Exhibit A4 of the Appendix), but a volatility sort gives a larger gap (2.3 points), and the crowding sort's gap is the smaller one in seven of nine episodes (difference of 0.4 points, t of 2.8). Even where crowded positions should matter most, the standard score adds nothing to volatility as a warning of drawdowns. Two features of these episodes differ from recessions. First, the most crowded factors moved together: their average pairwise correlation was 0.25, against 0.07 for the least crowded (t of 3.2 for the difference; Exhibit A4), whereas in recessions the two groups are equally correlated. A volatility sort shows the same pattern,

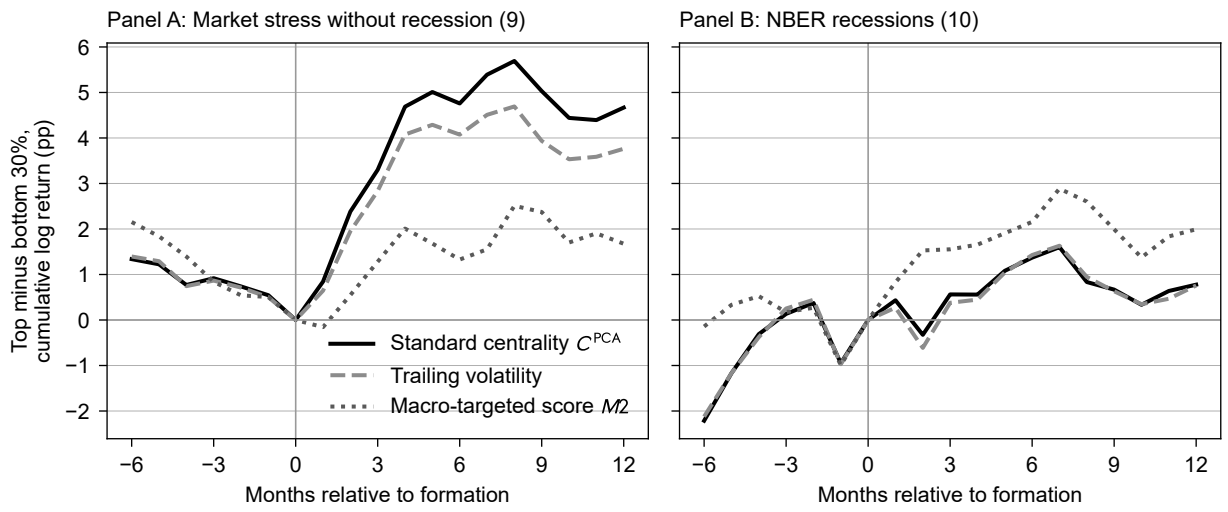


Exhibit 5: **Crowded Factors around Market Stress and Recessions.** JKP US factors. Cumulative mean log return of the factors in the top 30% minus the bottom 30% of each signal, formed at the market peak (Panel A) or in the month before the NBER peak (Panel B), from six months before to twelve months after formation, averaged across events. Panel A: the nine US market drawdowns of at least 15% in 1957–2024 that do not overlap an NBER recession. Panel B: the ten NBER recessions of Exhibit 4. Monthly data cannot resolve episodes that unfold within a month, such as the quantitative-equity crisis of August 2007.

slightly weaker (a correlation gap of 0.13, against 0.17). Second, the deeper drawdowns of individual crowded factors were not losses of the crowded group as a whole: from the market peak, the most crowded group earned 4.7 percentage points more than the least crowded over the following year (Panel A), and the most volatile group 3.8 points more. Reducing crowded sleeves at the market peak would not have helped on average. The macro-targeted score again does not identify the factors that lose most (its top group fell less, by 0.8 points).

5 Crowding in Portfolio Decisions

To test whether crowding is a return signal, Exhibit 6 tilts the equal-weighted factor portfolio toward factors whose score is high relative to its own recent history, with a long-only rank-weighting rule: each factor's score is ranked within its trailing 12 months and these percentiles are ranked across factors; the factors in the top and bottom 30% of the ranking receive weight $1+0.5\,z$, with z the centered cross-sectional rank, and the weights are renormalized (see Kelly,

Pruitt, and Su, 2019, for the rank transformation). The 30% tails, the tilt of 0.5, and the half weight of the crowding cap below are the choices of the development sample and were not tuned on the JKP data.

	Active		On equal-weighted		Alpha of active return after		
Tilt toward	return (%)	IR	β	α (%)	Volatility	Persistence	Both
Panel A: US, 1958–2024							
C^{PCA}	−0.01	−0.04	1.05	−0.13 (−4.8)	0.00 (0.1)	−0.02 (−0.7)	0.00 (−0.1)
C^{COR}	0.00	0.02	1.00	0.01 (0.2)	0.00 (0.2)	−0.02 (−1.0)	−0.02 (−0.8)
$M2$	0.02	0.11	0.99	0.06 (1.8)	0.02 (0.8)	0.02 (0.9)	0.02 (0.8)
Volatility tilt	−0.01	−0.07	1.03	−0.08 (−3.2)			
Premium persistence tilt	0.30**	0.50	1.00	0.30 (2.2)			
Panel B: Developed ex US, 1991–2024							
C^{PCA}	−0.02	−0.10	1.07	−0.16 (−4.7)	0.02 (0.8)	−0.01 (−0.3)	0.02 (0.6)
C^{COR}	0.00	0.01	1.01	−0.02 (−0.7)	0.01 (0.3)	−0.02 (−0.8)	−0.01 (−0.5)
$M2$	−0.02	−0.12	1.00	−0.02 (−0.4)	−0.03 (−0.9)	−0.06 (−1.8)	−0.07 (−2.1)
Volatility tilt	−0.03	−0.22	1.04	−0.12 (−4.4)			
Premium persistence tilt	0.24**	0.63	1.02	0.20 (1.6)			

Exhibit 6: **Is Crowding a Return Signal?** JKP factors (capped value weights). Long-only tilts with the portfolio rule of the article toward each signal, relative to the equal-weighted portfolio of the factors with a signal. Active return: annualized mean of the tilt minus the equal-weighted portfolio; stars from a one-sided stationary block bootstrap (5,000 draws, mean block 3 months); * and ** denote significance at the 5% and 1% levels. IR: information ratio. β and α: slope and annualized intercept of the tilt's return on the equal-weighted portfolio. Alpha after: intercept of the active return regressed on the active returns of the volatility tilt (trailing 60-month standard deviation), the premium-persistence tilt (trailing 60-month mean), or both, built with the same rule. Newey–West t in parentheses. Common months of all tilts.

No crowding tilt earns a significant active return, in the United States or in developed markets. Tilts toward standard crowding raise the portfolio's exposure to the equal-weighted factor portfolio (β of 1.05) and earn less than that exposure implies: an alpha of −0.13% a year (t of −4.8) in the United States and −0.16% (t of −4.7) in developed markets. The tilt toward volatile factors behaves the same way, which is what a volatility ranking predicts. Tilts toward correlation-based centrality and toward $M2$ earn active returns close to zero; after controlling for the volatility and premium-persistence tilts, their alphas are insignificant in the United States and, for $M2$ in developed markets, slightly negative (−0.07% a year, t of −2.1). Only the persistence of factor premia earns an active return (0.30% a year in the United States), in line with the evidence on factor momentum (Gupta and Kelly, 2019; Ehsani and Linnainmaa, 2022). We find no evidence that crowding scores capture mispricing;

their information concerns risk.

For the sizing of factor sleeves, Exhibit 7 compares equal weights, inverse-volatility weights, and long-only equal risk contribution (Maillard, Roncalli, and Teïletche, 2010), each with and without halving the weight of the 30% most crowded factors.

Panel A: US, 1957–2024 Portfolio	Vol. (%)	Sharpe	ΔSharpe	Max DD (%)	NBER DD mean / worst (%)
Equal weights	3.47	0.77		−18.71	−1.57 / −3.60
Equal weights, cap on C^{PCA}	2.93	0.90	0.13**	−14.55	−1.33 / −3.28
Equal weights, cap on C^{COR}	3.23	0.78	0.01	−17.36	−1.49 / −4.32
Inverse volatility	2.28	1.05	0.29**	−9.93	−1.19 / −2.83
Inverse volatility, cap on C^{PCA}	2.04	1.17	0.11**	−7.82	−1.17 / −2.75
Inverse volatility, cap on C^{COR}	2.08	1.09	0.04*	−8.84	−1.17 / −3.18
Equal risk contribution	0.62	1.30	0.25	−1.77	−0.32 / −0.70
Equal risk contribution, cap on C^{PCA}	0.78	1.13	−0.17*	−3.30	−0.40 / −0.72
Equal risk contribution, cap on C^{COR}	0.76	1.04	−0.26*	−3.35	−0.39 / −0.79
Panel B: Developed ex US, 1991–2024 Portfolio	Vol. (%)	Sharpe	ΔSharpe	Max DD (%)	NBER DD mean / worst (%)
Equal weights	1.97	1.06		−7.95	−1.61 / −2.28
Equal weights, cap on C^{PCA}	1.68	1.18	0.12**	−5.93	−1.26 / −1.92
Equal weights, cap on C^{COR}	1.81	1.08	0.02	−7.08	−1.29 / −1.93
Inverse volatility	1.45	1.23	0.18**	−5.25	−1.10 / −1.73
Inverse volatility, cap on C^{PCA}	1.32	1.29	0.06	−4.05	−0.95 / −1.61
Inverse volatility, cap on C^{COR}	1.37	1.23	0.00	−4.82	−0.93 / −1.63
Equal risk contribution	0.54	1.00	−0.24	−2.38	−0.22 / −0.36
Equal risk contribution, cap on C^{PCA}	0.58	1.08	0.08	−1.70	−0.22 / −0.35
Equal risk contribution, cap on C^{COR}	0.63	0.78	−0.22	−2.78	−0.27 / −0.58

Exhibit 7: **Sizing Factor Sleeves: Risk-Based Weights and Crowding Caps.** JKP factors (capped value weights) with a complete 60-month history; weights formed at the end of month $t-1$ and held over t. Inverse volatility: weights proportional to the inverse trailing 60-month volatility; equal risk contribution: long-only weights with equal contributions to the variance from the trailing 60-month sample covariance. Cap: the 30% of factors with the highest centrality (C^{PCA} from covariances, C^{COR} from correlations) at half weight, renormalized. Sharpe ratios annualized, without risk-free rate (zero-investment returns), gross of costs. ΔSharpe: difference to the base portfolio (equal weights for inverse volatility and for capped equal weights; inverse volatility for equal risk contribution and capped inverse volatility; equal risk contribution for capped equal risk contribution); * and ** denote significance at the 5% and 1% levels in the Ledoit and Wolf (2008) HAC test (Bartlett kernel, Newey–West lags), Holm-adjusted for the two caps on equal risk contribution. Max DD: maximum drawdown over the sample; NBER DD: within-recession maximum drawdown, mean and worst across NBER recessions that start after the first formation month.

For an investor who holds the factors in equal weights, capping the most crowded factors helps: in the United States it raises the Sharpe ratio from 0.77 to 0.90 and lowers volatility, because the cap removes weight from the most volatile factors. Inverse-volatility weights do more (1.05), and equal risk contribution, which accounts for correlations, does most (1.30,

although its gain over inverse-volatility weights is not significant) and has the shallowest recession drawdowns. On top of inverse-volatility weights, which ignore correlations, the standard cap still helps (1.17); on top of equal risk contribution it lowers the Sharpe ratio (to 1.13, Holm-adjusted p of 0.04), and the cap on correlation-based crowding lowers it further (1.04, p of 0.02). In developed markets, equal risk contribution does not beat inverse-volatility weights (1.00 against 1.23), the cap's gain on inverse-volatility weights is not significant (1.29, +0.06), and no cap improves equal risk contribution. Crowding caps therefore act as a partial substitute for risk budgeting in portfolios that ignore correlations and add nothing to equal risk contribution; which risk-based rule works best differs between the regions. Sharpe ratios are gross of costs and compare portfolios rather than estimate attainable performance; volatility-based rules are not uniformly reliable either (Bongaerts, Kang, and van Dijk, 2020).

The results assign each tool one job (Exhibit 8). Risk-based weights set the size of each factor sleeve. Standard crowding scores restate volatility and add nothing to that job. The macro-targeted score, used alongside volatility and the cycle beta, shows which sleeves carry the portfolio's business-cycle exposure and is a natural input to scenario design and to the monitoring of how the composition of cyclical risk changes. None of the scores is a profitable return signal.

In practice, the scores require only factor returns and a business-cycle indicator, and they can be computed each month with standard tools.

The inputs are a panel of monthly long–short factor returns, either an investor's own factor sleeves or public factors such as those of Jensen et al. (2023); at least five years of history per factor; and a real-time business-cycle indicator that combines valuation, credit, activity, and inflation information, such as that of Swade et al. (2024). Publication lags of the macroeconomic inputs should be respected; we leave out the last three months.

Each month, the calculation takes four steps. (1) Compute standard centrality from the

Measure	Use it for	Evidence (United States unless stated)	Do not use it for
Volatility and risk-based weights	Sizing factor sleeves	Inverse volatility raises the Sharpe ratio from 0.77 to 1.05; equal risk contribution to 1.30, not significantly above inverse volatility (Exhibit 7)	—
Standard centrality C^{PCA}	Reading as a volatility report; a cap only in portfolios that ignore correlations	Rank correlation with volatility 0.96; no information beyond the controls (-0.11, t of -1.8) (Exhibit 3)	Positioning; return forecasts (alpha -0.13%, t of -4.8); drawdown warnings beyond volatility (Exhibit A4); caps on equal risk contribution (Sharpe 1.30 to 1.13)
Correlation-based centrality C^{COR}	Locating shared exposure free of scale	Rank correlation with volatility 0.57; no information about future cycle exposure beyond the controls (Exhibit 3)	Cycle exposure; return forecasts
Macro-targeted score $M2$	Monitoring where cyclical exposure is concentrated; choosing sleeves for a downturn scenario	Adds to volatility and the cycle beta for future cycle sensitivity (pooled 0.078, t of 3.1); top-$M2$ factors move together in recessions (correlation 0.22 against 0.04) (Exhibits 3 and 4)	Ranking recession losses (drawdown difference 0.34, t of 0.4); sizing; return forecasts
Signed cycle loadings λ	Direction of each sleeve's exposure in a stress scenario	The size of the long-window cycle beta predicts future cycle sensitivity (pooled 0.066, t of 2.9; Exhibit 3); its sign is not tested	Sizing

Exhibit 8: **Which Measure for Which Decision.** Summary of Exhibits 3 to 7. Incremental coefficients are rank-regression coefficients with Newey–West t-statistics; Sharpe ratios are gross of costs. In developed markets outside the United States, equal risk contribution does not beat inverse-volatility weights (Sharpe ratio 1.00 against 1.23).

covariance matrix of the trailing 60 months (Equation 1) and report it next to each factor's trailing volatility; a rank correlation near one is the signal that the crowding report restates the risk report. (2) Estimate the innovation of the business-cycle indicator and each factor's loading on it, using all available history. (3) Demean the trailing returns, apply the targeted operator of Equation (2), and compute $M2$ (Equation 3). (4) Read the $M2$ ranking next to the rankings by volatility and long-window cycle beta: a high $M2$ flags factors whose

comovement is tied to the business cycle beyond what their volatility and cycle beta show.

The macro-targeted score belongs in risk monitoring and scenario design, not in position sizing or return forecasting. A risk committee can track which sleeves carry the portfolio's cyclical exposure and how the ranking changes from month to month, and can test a business-cycle scenario on the top-ranked sleeves. Because crowded sleeves can sit on opposite sides of the same common component, the scenario should use the signed loadings of the sleeves rather than treat them as one position. A scenario for a market sell-off without a recession should instead assume that the most volatile sleeves move together, as they did on average in the nine episodes since 1957; the standard crowding score selects nearly the same sleeves as volatility. Position sizes, before costs, are better set by risk-based weights, which already account for the volatility that standard crowding scores restate. A crowding cap is a reasonable shortcut only for portfolios that ignore correlations, mainly those held in equal weights, where it removes weight from the most volatile sleeves.

6 Robustness and Discussion

Exhibit 9 repeats the main tests in two alternative factor universes, both specified in advance. The first removes the exclusion rule and uses all 153 factors, including momentum, reversal, size, and liquidity factors. The second removes from the main universe the 18 factors whose sorting variable is a statistic of past stock returns: betas, idiosyncratic and total volatility, maximum returns, market correlation, and skewness. Most of them (11 of 18) belong to the low-risk cluster of Jensen et al. (2023), the theme that a crowding map flags as the most crowded in nearly every month (Exhibit 2).

The volatility content of standard centrality does not depend on the universe: its rank correlation with volatility is 0.94 with all 153 factors and unchanged at 0.96 without the return-based risk factors, and in both universes the recession drawdowns of crowded factors

Statistic	122 factors (main)	104 factors	153 factors
Rank correlation of C^{PCA} with volatility, US	0.961	0.961	0.942
Rank correlation of C^{PCA} with volatility, developed ex US	0.945	0.945	0.940
NBER max. drawdown difference, sorted on C^{PCA}, US (pp)	−6.59 (−8.1)	−5.23 (−8.0)	−6.52 (−7.4)
NBER max. drawdown difference, sorted on volatility, US (pp)	−6.84 (−8.1)	−5.60 (−9.0)	−7.16 (−6.6)
$M2$, future $\lvert\beta(\Delta Z)\rvert$, US 1968–2024	0.084 (3.3)	0.039 (1.9)	0.088 (3.8)
$M2$, future $\lvert\beta(\Delta Z)\rvert$, developed ex US	0.052 (2.4)	0.062 (2.3)	0.044 (1.9)
$M2$, future $\lvert\beta(\Delta Z)\rvert$, pooled	0.078 (3.1)	0.060 (2.6)	0.074 (3.0)
$M2$, future $R^2(\Delta Z)$, pooled	0.057 (2.3)	0.045 (2.1)	0.055 (2.1)
$M2$ beyond $\lvert\lambda\rvert$, future $\lvert\beta(\Delta Z)\rvert$, pooled	0.086 (2.6)	0.055 (1.7)	0.082 (2.7)
$M2$ beyond $\lvert\lambda\rvert$, future $R^2(\Delta Z)$, pooled	0.096 (2.9)	0.073 (2.1)	0.094 (2.6)

Exhibit 9: **Robustness of the Main Results to the Factor Universe.** Pre-registered robustness checks. 122 factors: main universe (Exhibits 3 and 4). 104 factors: without the 18 factors whose sorting variable is computed from past stock returns (market, Dimson, Frazzini–Pedersen and downside betas; idiosyncratic and total volatility; maximum returns; market correlation; coskewness and skewness); accounting-based risk measures are kept. 153 factors: all JKP factors, without the exclusion rule. Designs of Exhibits 3 and 4; incremental coefficients with the controls of Exhibit 3 including C^{PCA}; Newey–West t (24 lags), or t across recessions, in parentheses.

match those of volatile factors. The macro-targeted score keeps its information with all 153 factors (pooled t of 3.0, and 2.7 beyond the long-window cycle beta). Without the return-based risk factors it is weaker: pooled it remains significant (t of 2.6), but in the United States alone it is at the margin of significance (t of 1.9), and beyond the long-window beta it holds for the cycle's share of variance (t of 2.1) but not for cycle sensitivity (t of 1.7). Part of the cyclical information that the target recovers therefore comes from factors that sort stocks on their market and volatility risk, plausibly because their exposure varies with the price of risk over the business cycle.

In emerging markets, standard centrality is again a volatility ranking (rank correlation of 0.90), and the incremental information of the macro-targeted score is positive but not significant (0.035, t of 1.0); at the level of the nine JKP themes, the volatility content is the same (0.96), and, without controls, the macro-targeted score is related to the themes' future cycle sensitivity (0.21, t of 4.8; Exhibit A5 of the Appendix). The emerging-market sample starts in 1994, contains three US recessions, and uses a US business-cycle indicator, so it has little power for the second contribution.

In the anomaly themes of Chen and Zimmermann (2022), on which the analysis was

developed, five results differ. Standard centrality is significant beyond the controls there (0.27, t of 3.1, from 1990 onward and with a slightly different control set; Supplemental Material), whereas on the JKP factors it is not. Regime-dependent risk-adjusted performance (Exhibit 1) was not distinguishable from chance there (3 of the 24 themes with Fama–French factor data). Correlation-based centrality carried information about the cycle's future share of variance beyond volatility there, but only after 1990; it does not replicate on the JKP factors. The macro-targeted score is significant there as well (coefficient of 0.081, t of 3.0), but not beyond the long-window cycle beta (t of 1.4), so the evidence that the targeted decomposition adds information beyond that beta rests on the JKP data. And a crowding cap did not improve inverse-volatility weights across the 27 themes, whereas it does across the 122 US JKP factors, where many factors with overlapping characteristics share risk that inverse-volatility weights ignore. The volatility content of standard centrality (rank correlation of 0.92) and its recession record are the same in both data sets.

Why does a score designed to measure crowding end up measuring volatility? Its construction favors volatile series (Section 2), and the data make this tendency decisive. Across factor portfolios the dispersion of volatility is large, from about 2% to 14% a year among the JKP themes, and dominates the dispersion of correlations, so the score sorts factors by their volatility before it sorts them by how much of their risk is common. Comovement alone is not a cyclical signal either; it becomes one when the decomposition is directed at the business cycle.

The authors of the centrality score read high volatility as the footprint of crowded trading (Kinlaw et al., 2019). The positioning literature suggests that the causality more often runs the other way: capital leaves factor strategies when their volatility rises (Hanson and Sunderam, 2014; Barroso et al., 2022). A score dominated by volatility therefore cannot separate a factor that is volatile because many investors trade it from a factor that is volatile because its firms are risky. Our results do not contradict crowding as an economic phenomenon; they show that the standard returns-based score does not isolate it. Holdings and flows remain

the more direct measures of positioning, and returns-based scores are best read as measures of shared risk.

A macroeconomic target moves the decomposition only if it is informative about how factors load on the business cycle (Bryzgalova et al., 2026); a single cycle innovation with loadings estimated over a long history is such a target, and centering removes the premium channel. The resulting score is still noisy: its information is incremental, moderate in size, and weaker without the return-based risk factors.

The analysis has limitations. The scores are returns-based and say nothing directly about who holds a factor. The business-cycle indicator is a US indicator applied to all regions, so the developed-market evidence concerns exposure to the US, and largely global, cycle; regional indicators would add independent episodes. The out-of-sample evidence for the macro-targeted score covers 32 years of monthly cross-sections (385) and three recessions outside the United States. The factors are published anomalies, selected after the fact; their premia tend to be lower after publication (McLean and Pontiff, 2016), although the results here concern risk measurement more than premia. The factors are long–short portfolios with capped value weights, closer to investable strategies than equal-weighted academic portfolios but still before costs.

7 Conclusion

The returns-based crowding scores that factor investors monitor are, in their standard form, volatility rankings: across 122 factors in the United States and in developed markets, standard centrality ranks factors as volatility does, and the drawdowns of crowded factors in recessions and market sell-offs are those of volatile factors. Directed at a clean business-cycle innovation, the same decomposition becomes informative: a centered, macro-targeted score adds information about which factors will be most exposed to the business cycle, be-

yond volatility and standard crowding in the United States and out of sample, and beyond a long-window cycle beta in the pooled sample. Neither score is a profitable return signal, and crowding caps add nothing to risk budgets that account for correlations.

For an institutional factor investor, this suggests three changes. Read a standard crowding score as a volatility report, not as evidence of crowded positions. Size factor sleeves with risk-based weights; a crowding cap is at best a substitute for them. Monitor business-cycle concentration with a score directed at the business cycle, read next to volatility and cycle betas, and describe the crowded set by the signed exposures of its members when designing stress scenarios.

Two extensions would sharpen these conclusions. Regional business-cycle indicators would add independent downturns outside the United States and test whether the macro-targeted score captures local or global cyclical risk. And combining returns-based scores with holdings or flow data would show when common risk in factor returns reflects common positions.

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Declaration of generative AI use. The author used a generative AI assistant (Claude, Anthropic) for programming support in the data-processing and analysis code, for literature searches, and for language editing of the manuscript. The research questions, the design and pre-registration of the tests, and the interpretation of the results are the author's; all code, results, citations, and text were checked by the author, who takes full responsibility for the content of this article.

Data availability. Factor returns are publicly available from Jensen, Kelly, and Pedersen

(https://jkpfactors.com); anomaly returns of the development sample from the Open Source Asset Pricing project (https://www.openassetpricing.com); the Fama–French factors from Kenneth French's data library (https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html); the inputs to the business-cycle indicator from Robert Shiller's online data (http://www.econ.yale.edu/~shiller/data.htm) and the Federal Reserve Bank of St. Louis (FRED, https://fred.stlouisfed.org); and the recession dates from the NBER (https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions). A replication package reproduces every exhibit from these data and is available from the author on request.

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Appendix

This Appendix documents the data and the estimators and reports the evidence that the article cites but does not show. Section A describes the JKP factors, Section B the estimators, Section C the recession evidence event by event, Section D the emerging-market and theme-level results, and Section E the original target-PCA design. The development sample of anomaly themes on which the analysis was first developed is documented in the Supplemental Material. Exhibits carry the prefix A.

A. The JKP Factors

The factors are the capped value-weighted long–short factors of Jensen et al. (2023), downloaded from `https://jkpfactors.com` (version of August 2026, data through December 2025; the analysis ends in December 2024). Each factor goes long the top tercile and short the bottom tercile of a characteristic, with breakpoints from stocks larger than the 20th NYSE size percentile and stock weights equal to market capitalization capped at the 80th NYSE percentile; factors are signed as in the original studies. The regional factors aggregate country factors. The block *developed ex US* is the JKP region of developed markets, which excludes the United States (22 countries), and the block *emerging* the JKP region of emerging markets. Exhibit A1 describes the nine themes used for interpretation. The themes differ widely in risk: in the United States, the annualized volatility of the theme returns ranges from 2.4% (debt issuance) to 13.5% (low risk), and the low-risk and value themes are the two most volatile, which is why a covariance-based crowding score ranks them first (Exhibit 2 of the article). Their Sharpe ratios range from 0.02 (low leverage) to 1.07 (debt issuance). Exhibit A2 lists all 153 factors with their cluster and status: 122 are kept; 31 are excluded because their sorting variable is built from past returns (momentum 8, reversal 2, seasonality 10), from size or the price level (2), or from trading activity (9).

		US, 1957–2024			Developed ex US		Emerging	
Theme	Factors	Mean	Vol.	Sharpe	Mean	Sharpe	Mean	Sharpe
Debt issuance	7	2.56	2.39	1.07	1.26	0.40	0.98	0.34
Accruals	5	3.48	4.30	0.81	1.17	0.33	2.07	0.54
Quality	17	3.47	5.29	0.66	2.16	0.42	2.74	0.47
Profit growth	10	1.59	3.31	0.48	1.53	0.45	2.37	0.87
Investment	20	3.30	7.09	0.47	2.01	0.43	0.89	0.23
Profitability	9	3.11	7.35	0.42	2.57	0.48	2.76	0.54
Value	18	3.95	10.76	0.37	4.50	0.68	4.16	0.73
Low risk	13	2.40	13.50	0.18	3.03	0.36	2.78	0.41
Low leverage	10	0.16	9.59	0.02	−1.10	−0.18	−2.69	−0.38

Exhibit A1: **JKP Themes.** The nine JKP clusters used for interpretation, each the equal-weighted mean of its kept factors (capped value-weighted long–short factors of Jensen, Kelly, and Pedersen 2023). Factors: number of kept factors (122 in total; 13 kept factors belong to the four excluded clusters and enter only the factor-level tests). Mean and volatility in % p.a., Sharpe ratio without risk-free rate. US: January 1957–December 2024; developed ex US from 1986-01 to 1986-02, emerging from 1988-08 to 1993-05 depending on the theme, to December 2024.

Exhibit A2: **JKP Factors and the Exclusion Rule.** The 153 factors of Jensen, Kelly, and Pedersen (2023): the 122 kept factors by JKP cluster, and the 31 excluded factors by the reason for their exclusion (sorting variable built from past returns, from the price level or size, or from trading activity); return-based risk measures (betas, volatilities, skewness) are kept. Description and original study as in the JKP factor documentation; for 11 factors that the documentation lists without an original study, the entry refers to the replication list of Hou, Xue, and Zhang (2020). The original studies are given as identifiers and are not repeated in the References. Clusters are those of the JKP classification; the kept factors in the seasonality, short-term reversal, and size clusters are sorted on risk or fundamental characteristics, not on past returns or size, and are therefore kept under the exclusion rule.

Group, theme, and factor	Description	Original study
Kept factors (122)		
Accruals (5 factors)		
`cowc_gr1a`	Change in current operating working capital	Richardson et al. (2005)
`oaccruals_at`	Operating accruals	Sloan (1996)
`oaccruals_ni`	Percent operating accruals	Hafzalla Lundholm and Van Winkle (2011)
`taccruals_at`	Total accruals	Richardson et al. (2005)
`taccruals_ni`	Percent total accruals	Hafzalla Lundholm and Van Winkle (2011)
Debt issuance (7 factors)		
`capex_abn`	Abnormal corporate investment	Titman Wei and Xie (2004)
`debt_gr3`	Growth in book debt (3 years)	Lyandres Sun and Zhang (2008)

Group, theme, and factor	Description	Original study
fnl_gr1a	Change in financial liabilities	Richardson et al. (2005)
ncol_gr1a	Change in noncurrent operating liabilities	Richardson et al. (2005)
nfna_gr1a	Change in net financial assets	Richardson et al. (2005)
ni_ar1	Earnings persistence	Francis et al. (2004)
noa_at	Net operating assets	Hirshleifer et al. (2004)
Investment (20 factors)		
aliq_at	Liquidity of book assets	Ortiz-Molina and Phillips (2014)
at_gr1	Asset Growth	Cooper Gulen and Schill (2008)
be_gr1a	Change in common equity	Richardson et al. (2005)
capx_gr1	CAPEX growth (1 year)	Xie (2001)
capx_gr2	CAPEX growth (2 years)	Anderson and Garcia-Feijoo (2006)
capx_gr3	CAPEX growth (3 years)	Anderson and Garcia-Feijoo (2006)
coa_gr1a	Change in current operating assets	Richardson et al. (2005)
col_gr1a	Change in current operating liabilities	Richardson et al. (2005)
emp_gr1	Hiring rate	Belo Lin and Bazdresch (2014)
inv_gr1	Inventory growth	Belo and Lin (2011)
inv_gr1a	Inventory change	Thomas and Zhang (2002)
lnoa_gr1a	Change in long-term net operating assets	Fairfield Whisenant and Yohn (2003)
mispricing_mgmt	Mispricing factor: Management	Stambaugh and Yuan (2016)
ncoa_gr1a	Change in noncurrent operating assets	Richardson et al. (2005)
nncoa_gr1a	Change in net noncurrent operating assets	Richardson et al. (2005)
noa_gr1a	Change in net operating assets	Hirshleifer et al. (2004)
ppeinv_gr1a	Change PPE and Inventory	Lyandres Sun and Zhang (2008)
sale_gr1	Sales Growth (1 year)	Lakonishok Shleifer and Vishny (1994)
sale_gr3	Sales Growth (3 years)	Lakonishok Shleifer and Vishny (1994)
saleq_gr1	Sales growth (1 quarter)	Hou, Xue, and Zhang (2020), list
Low leverage (10 factors)		
age	Firm age	Jiang Lee and Zhang (2005)

Group, theme, and factor	Description	Original study
aliq_mat	Liquidity of market assets	Ortiz-Molina and Phillips (2014)
at_be	Book leverage	Fama and French (1992)
cash_at	Cash-to-assets	Palazzo (2012)
netdebt_me	Net debt-to-price	Penman Richardson and Tuna (2007)
ni_ivol	Earnings volatility	Francis et al. (2004)
rd5_at	R&D capital-to-book assets	Li (2011)
rd_sale	R&D-to-sales	Chan Lakonishok and Sougiannis (2001)
tangibility	Asset tangibility	Hahn and Lee (2009)
z_score	Altman Z-score	Dichev (1998)
Low risk (13 factors)		
beta_60m	Market Beta	Fama and MacBeth (1973)
beta_dimson_21d	Dimson beta	Dimson (1979)
betabab_1260d	Frazzini-Pedersen market beta	Frazzini and Pedersen (2014)
betadown_252d	Downside beta	Ang Chen and Xing (2006)
earnings_variability	Earnings variability	Francis et al. (2004)
ivol_capm_21d	Idiosyncratic volatility from the CAPM (21 days)	Hou, Xue, and Zhang (2020), list
ivol_capm_252d	Idiosyncratic volatility from the CAPM (252 days)	Ali Hwang and Trombley (2003)
ivol_ff3_21d	Idiosyncratic volatility from the Fama-French 3-factor model	Ang et al. (2006)
ivol_hxz4_21d	Idiosyncratic volatility from the q-factor model	Hou, Xue, and Zhang (2020), list
ocfq_saleq_std	Cash flow volatility	Huang (2009)
rmax1_21d	Maximum daily return	Bali Cakici and Whitelaw (2011)
rmax5_21d	Highest 5 days of return	Bali, Brown, Murray and Tang (2017)
rvol_21d	Return volatility	Ang et al. (2006)
Profit growth (10 factors)		
dsale_dinv	Change sales minus change Inventory	Abarbanell and Bushee (1998)
dsale_drec	Change sales minus change receivables	Abarbanell and Bushee (1998)
dsale_dsga	Change sales minus change SG&A	Abarbanell and Bushee (1998)

Group, theme, and factor	Description	Original study
niq_at_chg1	Change in quarterly return on assets	Hou, Xue, and Zhang (2020), list
niq_be_chg1	Change in quarterly return on equity	Hou, Xue, and Zhang (2020), list
niq_su	Standardized earnings surprise	Foster Olsen and Shevlin (1984)
ocf_at_chg1	Change in operating cash flow to assets	Bouchard, Krüger, Landier and Thesmar (2019)
sale_emp_gr1	Labor force efficiency	Abarbanell and Bushee (1998)
saleq_su	Standardized Revenue surprise	Jegadeesh and Livnat (2006)
tax_gr1a	Tax expense surprise	Thomas and Zhang (2011)
Profitability (9 factors)		
ebit_bev	Return on net operating assets	Soliman (2008)
ebit_sale	Profit margin	Soliman (2008)
f_score	Piotroski F-score	Piotroski (2000)
ni_be	Return on equity	Haugen and Baker (1996)
niq_be	Quarterly return on equity	Hou Xue and Zhang (2015)
o_score	Ohlson O-score	Dichev (1998)
ocf_at	Operating cash flow to assets	Bouchard, Krüger, Landier and Thesmar (2019)
ope_be	Operating profits-to-book equity	Fama and French (2015)
ope_bel1	Operating profits-to-lagged book equity	Hou, Xue, and Zhang (2020), list
Quality (17 factors)		
at_turnover	Capital turnover	Haugen and Baker (1996)
cop_at	Cash-based operating profits-to-book assets	Hou, Xue, and Zhang (2020), list
cop_atl1	Cash-based operating profits-to-lagged book assets	Ball et al. (2016)
dgp_dsale	Change gross margin minus change sales	Abarbanell and Bushee (1998)
gp_at	Gross profits-to-assets	Novy-Marx (2013)
gp_atl1	Gross profits-to-lagged assets	Hou, Xue, and Zhang (2020), list
mispricing_perf	Mispricing factor: Performance	Stambaugh and Yuan (2016)
ni_inc8q	Number of consecutive quarters with earnings increases	Barth Elliott and Finn (1999)
niq_at	Quarterly return on assets	Balakrishnan Bartov and Faurel (2010)

Group, theme, and factor	Description	Original study
op_at	Operating profits-to-book assets	Hou, Xue, and Zhang (2020), list
op_atl1	Operating profits-to-lagged book assets	Ball et al. (2016)
opex_at	Operating leverage	Novy-Marx (2011)
qmj	Quality minus Junk: Composite	Asness, Frazzini and Pedersen (2018)
qmj_growth	Quality minus Junk: Growth	Asness, Frazzini and Pedersen (2018)
qmj_prof	Quality minus Junk: Profitability	Asness, Frazzini and Pedersen (2018)
qmj_safety	Quality minus Junk: Safety	Asness, Frazzini and Pedersen (2018)
sale_bev	Asset turnover	Soliman (2008)
Seasonality (7 factors)		
corr_1260d	Market correlation	Asness, Frazzini, Gormsen, Pedersen (2020)
coskew_21d	Coskewness	Harvey and Siddique (2000)
dbnetis_at	Net debt issuance	Bradshaw Richardson and Sloan (2006)
kz_index	Kaplan-Zingales index	Lamont Polk and Saa-Requejo (2001)
lti_gr1a	Change in long-term investments	Richardson et al. (2005)
pi_nix	Taxable income-to-book income	Lev and Nissim (2004)
sti_gr1a	Change in short-term investments	Richardson et al. (2005)
Short-term reversal (5 factors)		
iskew_capm_21d	Idiosyncratic skewness from the CAPM	Hou, Xue, and Zhang (2020), list
iskew_ff3_21d	Idiosyncratic skewness from the Fama-French 3-factor model	Bali Engle and Murray (2016)
iskew_hxz4_21d	Idiosyncratic skewness from the q-factor model	Hou, Xue, and Zhang (2020), list
rmax5_rvol_21d	Highest 5 days of return scaled by volatility	Asness, Frazzini, Gormsen, Pedersen (2020)
rskew_21d	Total skewness	Bali Engle and Murray (2016)
Size (1 factor)		
rd_me	R&D-to-market	Chan Lakonishok and Sougiannis (2001)
Value (18 factors)		
at_me	Assets-to-market	Fama and French (1992)

Group, theme, and factor	Description	Original study
be_me	Book-to-market equity	Rosenberg Reid and Lanstein (1985)
bev_mev	Book-to-market enterprise value	Penman Richardson and Tuna (2007)
chcsho_12m	Net stock issues	Pontiff and Woodgate (2008)
debt_me	Debt-to-market	Bhandari (1988)
div12m_me	Dividend yield	Litzenberger and Ramaswamy (1979)
ebitda_mev	EBITDA-to-market enterprise value	Loughran and Wellman (2011)
eq_dur	Equity duration	Dechow Sloan and Soliman (2004)
eqnetis_at	Net equity issuance	Bradshaw Richardson and Sloan (2006)
eqnpo_12m	Equity net payout	Daniel and Titman (2006)
eqnpo_me	Net payout yield	Boudoukh et al. (2007)
eqpo_me	Payout yield	Boudoukh et al. (2007)
fcf_me	Free cash flow-to-price	Lakonishok Shleifer and Vishny (1994)
ival_me	Intrinsic value-to-market	Frankel and Lee (1998)
netis_at	Net total issuance	Bradshaw Richardson and Sloan (2006)
ni_me	Earnings-to-price	Basu (1983)
ocf_me	Operating cash flow-to-market	Desai Rajgopal and Venkatachalam (2004)
sale_me	Sales-to-market	Barbee Mukherji and Raines (1996)
Excluded factors (31)		
Liquidity / trading activity (9 factors)		
ami_126d	Amihud Measure	Amihud (2002)
bidaskhl_21d	The high-low bid-ask spread	Corwin and Schultz (2012)
dolvol_126d	Dollar trading volume	Brennan Chordia and Subrahmanyam (1998)
dolvol_var_126d	Coefficient of variation for dollar trading volume	Chordia Subrahmanyam and Anshuman (2001)
turnover_126d	Share turnover	Datar Naik and Radcliffe (1998)
turnover_var_126d	Coefficient of variation for share turnover	Chordia Subrahmanyam and Anshuman (2001)

Group, theme, and factor	Description	Original study
zero_trades_126d	Number of zero trades with turnover as tiebreaker (6 months)	Liu (2006)
zero_trades_21d	Number of zero trades with turnover as tiebreaker (1 month)	Liu (2006)
zero_trades_252d	Number of zero trades with turnover as tiebreaker (12 months)	Liu (2006)
Past returns (momentum) (8 factors)		
prc_highprc_252d	Current price to high price over last year	George and Hwang (2004)
resff3_12_1	Residual momentum t-12 to t-1	Blitz Huij and Martens (2011)
resff3_6_1	Residual momentum t-6 to t-1	Blitz Huij and Martens (2011)
ret_12_1	Price momentum t-12 to t-1	Jegadeesh and Titman (1993)
ret_12_7	Price momentum t-12 to t-7	Novy-Marx (2012)
ret_3_1	Price momentum t-3 to t-1	Jegadeesh and Titman (1993)
ret_6_1	Price momentum t-6 to t-1	Jegadeesh and Titman (1993)
ret_9_1	Price momentum t-9 to t-1	Jegadeesh and Titman (1993)
Reversal (2 factors)		
ret_1_0	Short-term reversal	Jegadeesh (1990)
ret_60_12	Long-term reversal	De Bondt and Thaler (1985)
Seasonality (10 factors)		
seas_11_15an	Years 11-15 lagged returns, annual	Heston and Sadka (2008)
seas_11_15na	Years 11-15 lagged returns, nonannual	Heston and Sadka (2008)
seas_16_20an	Years 16-20 lagged returns, annual	Heston and Sadka (2008)
seas_16_20na	Years 16-20 lagged returns, nonannual	Heston and Sadka (2008)
seas_1_1an	Year 1-lagged return, annual	Heston and Sadka (2008)
seas_1_1na	Year 1-lagged return, nonannual	Heston and Sadka (2008)
seas_2_5an	Years 2-5 lagged returns, annual	Heston and Sadka (2008)
seas_2_5na	Years 2-5 lagged returns, nonannual	Heston and Sadka (2008)
seas_6_10an	Years 6-10 lagged returns, annual	Heston and Sadka (2008)
seas_6_10na	Years 6-10 lagged returns, nonannual	Heston and Sadka (2008)
Size / price level (2 factors)		
market_equity	Market Equity	Banz (1981)
prc	Price per share	Miller and Scholes (1982)

B. The Estimators

For a symmetric operator Ω of the N_t factors with eigenpairs (σ_j^2, v^j), $\sigma_1^2 \geq \sigma_2^2 \geq \ldots$, centrality is Equation (1) of the article with $K = 5$. It is nonnegative, sums to one, and is invariant to the sign of the eigenvectors. Standard centrality C^{PCA} uses the sample covariance matrix of the trailing 60 months, correlation-based centrality C^{COR} the sample correlation matrix, and uncentered centrality $C^{\text{PCA}}_{\text{unc}}$ the second-moment matrix $R^\top R/(NT)$ of the raw returns.

Following Bryzgalova et al. (2026), a cross-sectional target $H \in \mathbb{R}^{N\times L}$ and a time-series target $D \in \mathbb{R}^{T\times J}$ define the projections $P_H = H(H^\top H)^{-1}H^\top$ and $P_D = D(D^\top D)^{-1}D^\top$. The joint estimator applies principal component analysis to

$$\Omega \;=\; \frac{1}{NT}\left(I_N + \tilde{\gamma}_{XS}P_H\right) R^\top\left(I_T + \gamma_{TS}P_D\right)R\left(I_N + \tilde{\gamma}_{XS}P_H\right), \qquad \tilde{\gamma}_{XS} = \sqrt{\gamma_{XS}N/L + 1} - 1, \tag{A1}$$

which reduces to the second-moment matrix of R without targets ($\gamma_{XS} = \gamma_{TS} = 0$); with $\gamma_{TS} = 0$ or $\gamma_{XS} = 0$ it gives the cross-sectional or the time-series estimator. The operator is the solution of the penalized least-squares problem of Bryzgalova et al. (2026), whose time-series alpha target is their Equation (10).

The macro-targeted score of the article uses Equation (A1) with the following choices; everything is computed each month t from information available at t.

1. *Centering.* R is the matrix of trailing 60-month returns demeaned within the window, so the operator contains no average returns.
2. *Cross-sectional target ($L = 1$).* The cycle innovation u_s is the residual of a first-order autoregression of the monthly change ΔZ_s of the real-time business-cycle indicator, estimated by least squares on all months up to $t - 3$. Each factor's loading λ_i is the slope of its returns on u over all months up to $t-3$ (at least 36), and $H = (\lambda_1, \ldots, \lambda_N)^\top$.
3. *Time-series target.* D is the demeaned series $\min(\Delta_{12}Z_s, 0)$ of the negative part of the 12-month change of the indicator within the window; the last three months are excluded, the publication lag of the macroeconomic inputs.
4. *Strengths.* $(\gamma_{XS}, \gamma_{TS}) = (0.1, 10)$, the values of the original design, not tuned.

With $C^{\text{XS-TS},c}$ the resulting centrality, the macro-targeted shift is $M2 = C^{\text{XS-TS},c} - C^{\text{PCA}}$; it sums to zero across factors at every date. Without targets, the centered operator reproduces C^{PCA} exactly. The long-window cycle beta used as a control is $|\lambda_i|$.

The original estimator of the development sample uses the raw (uncentered) returns, a cross-sectional target $H = \Lambda$ of loadings of the factors on the monthly changes of the five components of the indicator ($L = 5$), estimated on the 60-month window without its last three months, and a time-series target D equal to the demeaned 12-month change of the indicator. Its excess centrality relative to standard centrality, $\Delta C^m = C^m - C^{\mathrm{PCA}}$, splits into a macroeconomic component $M^m = C^m - C^{\mathrm{PCA}}_{\mathrm{unc}}$ and an average-return component $G = C^{\mathrm{PCA}}_{\mathrm{unc}} - C^{\mathrm{PCA}}$ (Section E).

Future cycle sensitivity $|\beta(\Delta Z)|$ and the cycle's share of variance $R^2(\Delta Z)$ are the absolute slope and the R^2 of a regression of each factor's returns on ΔZ over months $t+1$ to $t+24$. Each month, the outcome's percentile rank is regressed on the percentile ranks of the signal and the controls (trailing 60-month volatility; absolute beta on the regional market over 60 months; trailing $|\beta(\Delta Z)|$ over the window without its last three months; the norm $\|\Lambda\|$ of the loadings on the monthly changes of the five indicator components; and, where stated, C^{PCA} and $|\lambda|$). The coefficients are averaged over time with Newey–West standard errors with 24 lags. Pooled tests demean the ranks within each block and stack the blocks month by month.

C. Recessions and Market Stress by Event

Exhibit A3 shows, for each US recession since 1957, the average within-recession maximum drawdown of the most and least crowded factors (Panel A) and of the most and least volatile factors (Panel B); Exhibit 4 of the article reports the averages. The most crowded factors fell more than the least crowded in every recession, by between 2.5 percentage points (1960) and 10.5 points (2001). The gap between the most and least volatile factors is almost the same in every episode, between 2.9 and 10.6 points; it differs by more than one percentage point only in 2007–09 (8.5 points for crowding, 10.1 for volatility). With a rank correlation of 0.96 between standard centrality and volatility, the two sorts select largely the same factors before each recession.

Exhibit A4 reports the nine US market drawdowns of at least 15% since 1957 that do not overlap an NBER recession (Exhibit 5 of the article). The episodes were selected by a rule fixed before the analysis: an episode runs from the market index's last peak before the drawdown from that peak first reaches −15% to the trough before the previous peak is regained. The rule yields 16 episodes since 1957, seven of which overlap a recession. In the nine remaining episodes, the drawdown gap between the most and least crowded factors ranges from −6.2 percentage points (2021–22) to +1.0 point (2011), and it is never more

than 0.1 point deeper than the gap of a volatility sort. In 2011 and 2018 the most crowded factors fell less than the least crowded.

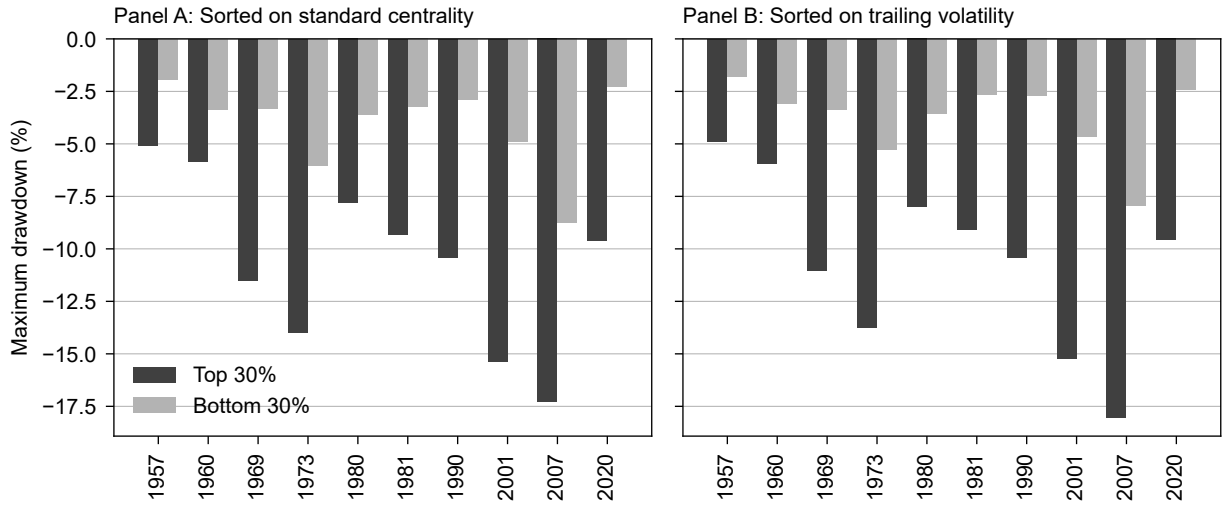


Exhibit A3: **Maximum Drawdown of Crowded and Volatile Factors in Each US Recession.** JKP US factors. Average within-recession maximum drawdown of the factors in the top and bottom 30% of standard centrality (Panel A) and of trailing 60-month volatility (Panel B) in the month before each NBER recession, 1957–2024.

		Drawdown gap, top minus bottom 30% (pp)				Correlation gap	
Episode (peak–trough)	Market drawdown (%)	C^{PCA}	Volatility	Difference	$M2$	C^{PCA}	Volatility
1961-11–1962-06	−26.0	−1.9	−2.3	0.4	1.5	−0.01	−0.05
1966-01–1966-09	−18.4	−4.9	−4.8	−0.1	−1.6	−0.16	−0.20
1983-06–1984-07	−23.2	−2.0	−2.5	0.5	2.1	0.19	0.19
1987-08–1987-11	−31.9	−1.1	−1.1	−0.1	0.3	0.25	0.22
1998-04–1998-08	−26.1	−2.0	−2.2	0.2	0.8	0.21	0.19
2011-04–2011-09	−22.0	1.0	0.0	1.1	0.5	0.41	0.28
2015-05–2016-01	−15.4	−0.7	−1.8	1.1	0.4	0.23	0.15
2018-08–2018-12	−17.8	0.5	0.1	0.4	1.3	0.26	0.19
2021-10–2022-09	−26.6	−6.2	−6.3	0.1	1.6	0.19	0.19
Mean, stress episodes (9)		−1.9 (−2.5)	−2.3 (−3.4)	0.4 (2.8)	0.8 (2.2)	0.17 (3.2)	0.13 (2.6)
Mean, NBER recessions (10)		−6.6 (−8.1)	−6.8 (−8.1)	0.3 (1.5)	0.3 (0.4)	0.00 (0.1)	0.01 (0.2)

Exhibit A4: **Crowded Factors in Market Stress Episodes.** JKP US factors. Episodes: US market drawdowns of at least 15% (peak to trough) in 1957–2024 that do not overlap an NBER recession, selected by a rule fixed before the analysis. Factors in the top and bottom 30% of each signal at the market peak; drawdown gap: mean within-episode maximum drawdown of the top group minus that of the bottom group, in percentage points. Difference: gap of the C^{PCA} sort minus gap of the volatility sort. Correlation gap: mean pairwise correlation of the monthly returns of the top group minus that of the bottom group during the episode. Mean rows: t-statistics across episodes in parentheses; the NBER row uses the recessions of Exhibit 4.

D. Emerging Markets and the Theme Level

Exhibit A5 repeats the measurement results for emerging-market factors and for the nine JKP themes in the United States (Section 6 of the article). In emerging markets, standard centrality has a rank correlation of 0.90 with trailing volatility (0.44 for correlation-based centrality). The drawdown gap between the most and least crowded factors in the three recessions since 1994 (−3.2 percentage points) matches that of a volatility sort (−3.3 points); with three events, neither gap is significant at the 5% level. On its own, the macro-targeted score is unrelated to future cycle sensitivity (−0.033, t of −1.3); beyond the controls, its coefficient is positive but not significant (0.035, t of 1.0). Inverse-volatility weights earn a higher Sharpe ratio than equal weights (1.14 versus 1.08), while equal risk contribution earns less (0.85), and a crowding cap does not change it (0.84). At the theme level, with nine themes and three principal components, the rank correlation of standard centrality with volatility is 0.96, correlation-based centrality is far less related to volatility (0.29), and the crowding and volatility sorts again produce almost the same recession drawdown gap (−4.6 and −4.5 points over the ten US recessions). Without controls, both standard centrality (0.43, t of 10.6) and the macro-targeted score (0.21, t of 4.8) are related to the themes' future cycle sensitivity; with nine themes, the panel is too small for the controlled regressions, so these statistics are descriptive.

The robustness of the main results to the factor universe, with all 153 factors and without the 18 return-based risk factors, is reported in Exhibit 9 of the article.

Statistic	Emerging, factors	US, 9 themes
Rank correlation C^{PCA} with volatility	0.905	0.959
Rank correlation C^{COR} with volatility	0.436	0.293
Spearman C^{PCA} $\lvert\beta(\Delta Z)\rvert$	0.218 (5.3)	0.431 (10.6)
Spearman C^{COR} $\lvert\beta(\Delta Z)\rvert$	0.129 (3.5)	0.126 (3.3)
Spearman $M2$ $\lvert\beta(\Delta Z)\rvert$	−0.033 (−1.3)	0.206 (4.8)
Incremental $M2$ $\lvert\beta(\Delta Z)\rvert$	0.035 (1.0)	
Spearman C^{PCA} $R^2(\Delta Z)$	−0.024 (−1.0)	0.013 (0.3)
Spearman C^{COR} $R^2(\Delta Z)$	0.001 (0.0)	−0.011 (−0.4)
Spearman $M2$ $R^2(\Delta Z)$	0.019 (0.8)	0.079 (1.8)
Incremental $M2$ $R^2(\Delta Z)$	0.029 (0.9)	
NBER max DD diff C^{PCA}	−3.20 (−1.9)	−4.62 (−3.9)
NBER max DD diff Volatility	−3.25 (−2.0)	−4.47 (−4.5)
Sharpe EW	1.08	
Sharpe Inverse volatility	1.14	
Sharpe ERC	0.85	
Sharpe ERC + cap C^{PCA}	0.84	

Exhibit A5: **Emerging Markets and the Theme Level (Descriptive).** Emerging: 122 JKP factors from the first month in which the region holds at least ten countries (formation from January 1994). US themes: the nine JKP themes, centrality with $K = 3$ components. Statistics as in Exhibits 3, 4, and 7; Newey–West t (24 lags) or t across recessions in parentheses; NBER drawdown differences (top minus bottom 30%) in percentage points, over 3 recessions (emerging, since 1994) and 10 recessions (US themes, since 1957). Sharpe ratios annualized.

E. The Original Target-PCA Design on JKP

Exhibit A6 applies the original, uncentered design (Section B) unchanged to the JKP US factors. The targets barely move the crowding map; the macroeconomic components carry no information about future cycle exposure; and the active returns of the targeted signals come from the average-return component G, which ranks factors partly by their past premia. The rank correlations of the targeted centralities with uncentered PCA centrality are 0.94 to 0.97, the macroeconomic components carry no information about future cycle sensitivity or the cycle's share of variance (t-statistics between -1.0 and 1.6), and among the components, only the average-return component G earns a significant active return (0.09% a year, significant at 1%); the excess centrality of the time-series estimator (0.07% a year, significant at 5%) inherits it, since its macroeconomic part is insignificant (0.04%). For comparison, a tilt on the trailing 60-month mean return earns 0.29% a year. The development sample showed the same pattern, and searches over the parameters, specified in advance, of the original design did not change it (Supplemental Material).

Panel A: Mean rank correlation with uncentered PCA centrality $C^{\text{PCA}}_{\text{unc}}$	C^{PCA}	C^{XS}	C^{TS}	$C^{\text{XS-TS}}$
Rank correlation	1.00	0.97	0.96	0.94
Panel B: Active return of tilts (% p.a.)				
Tilt on	Full sample	Recovery		
ΔC^{XS}	0.03	0.11		
M^{XS}	0.01	0.04		
ΔC^{TS}	0.07*	−0.03		
M^{TS}	0.04	−0.07		
$\Delta C^{\text{XS-TS}}$	0.04	0.04		
$M^{\text{XS-TS}}$	0.02	0.00		
G	0.09**	0.04		
Trailing 60m mean	0.29**	0.40**		
Panel C: Incremental information of M^m *for future cycle exposure, US 1968–2024*				
	$\lvert\beta(\Delta Z)\rvert$	$R^2(\Delta Z)$		
M^{XS}	0.016 (1.1)	0.027 (1.6)		
M^{TS}	−0.017 (−1.0)	−0.013 (−0.7)		
$M^{\text{XS-TS}}$	−0.001 (−0.1)	0.012 (0.7)		

Exhibit A6: **Uncentered Target PCA on JKP: Macroeconomic Target versus Average Returns.** JKP US factors, 1957–2024. Original target-PCA design of the development sample (uncentered second moment; XS target: loadings on the monthly changes of the five cycle components; TS target: comovement with the 12-month change of the indicator; $(\gamma_{XS}, \gamma_{TS}) = (0.1, 10)$). $\Delta C^m = C^m - C^{\text{PCA}} = M^m + G$ with $M^m = C^m - C^{\text{PCA}}_{\text{unc}}$ and $G = C^{\text{PCA}}_{\text{unc}} - C^{\text{PCA}}$. Panel B: long-only tilts with the article's rule; * and ** denote significance at the 5% and 1% levels (one-sided stationary block bootstrap); Recovery: months of the real-time indicator's Recovery regime. Panel C: design of Exhibit 3 (controls: volatility, absolute market beta, trailing $\lvert\beta(\Delta Z)\rvert$ with a three-month lag, $\|\Lambda\|$, C^{PCA}); Newey–West t (24 lags).